

LIFENG CHEN

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🎓 EDUCATION

University of Georgia

Ph.D. in Computer Science

Aug. 2026 – Present

Advisor: Tianming Liu

Beijing Jiaotong University

B.Sc. in Computer Science

Sep. 2022 – Jun. 2026

GPA: 3.9/4.0 Advisor: Xiaojie Jin

💡 RESEARCH INTERESTS

- Efficient Multimodal Generation
- Multimodal Foundation Models
- World Models

📄 PAPERS

1. **Detail++: Training-Free Detail Enhancer for Text-to-Image Diffusion Models.**
Lifeng Chen, Jiner Wang, Zihao Pan, et al.
Project page **IEEE Transactions on Image Processing** 2025
2. **ECHO: Efficient Chest X-ray Report Generation with One-step Block Diffusion.**
Lifeng Chen, Tianqi You*, Hao Liu, et al.*
Project page **ECCV 2026** 2026
3. **Genesis: Visual Augmentation Primitives for Multimodal Chain-of-Thought Reasoning.**
Fan Yang, Lifeng Chen*, Yipu Wang*, Han Wang*, Feilong Chen, Zhiyang Chen, Yidong Ma, et al.*
EMNLP 2026 Main Conference 2026
4. **Discrete Diffusion Models with MLLMs for Unified Medical Multimodal Generation.**
Jiawei Mao, Yuhan Wang, Lifeng Chen, et al.
Project page **ECCV 2026** 2026
5. **MedVIGIL: Evaluating Trustworthy Medical VLMs Under Broken Visual Evidence.**
Hanqi Jiang, Junhao Chen, Yi Pan, Lifeng Chen, et al.
Paper **NeurIPS 2026 (Evaluations & Datasets Track)** 2026

*Equal contribution (co-first authors).

🔬 RESEARCH EXPERIENCE

Xiaohongshu, Super Intelligence Team

Research Intern, Multimodal Foundation Models

Mar. 2026 – Aug. 2026

Advisor: Feilong Chen

- Developed **Genesis**, using GPT-Image-2 to generate answer-changing STEM training pairs through visual-primitive editing, with primitive-grounded reasoning traces. Trained MLLMs from **4B to 32B** using rubric-based multimodal CoT RL with primitive-level rewards along these traces; the 4B model outperformed 8B baselines, while the training pipeline raised an internal MLLM's ZEROBench_{sub} score from **23.12 to 32.33**.
- Proposed the **UNREAL Benchmark**, a multi-subject multimodal STEM benchmark that tests reasoning from visual evidence when scenes contradict commonsense and textbook priors.
- Co-designed **CoAnchor**, injecting priors from a frozen geometry foundation model into cross-view attention and progressively training MLLMs to use point correspondences as explicit CoT evidence; outperformed GPT-5.2 by **9.8 points** across four multi-view benchmarks.

Midea Group, AI Research Center

Aug. 2025 – Mar. 2026

Research Intern, Medical Multimodal Foundation Models

Advisor: Hao Liu

- Led **ECHO**, converting a pretrained autoregressive MLLM into a block-diffusion model using only **2.2%** of its SFT corpus; reduced adaptation FLOPs by **72.3%** and achieved **5.1×** practical speedup with one-step-per-block decoding. To retain generation quality, developed trajectory-level distillation that combines token distributions from the teacher’s multi-step denoising trajectory into a joint target, preserving cross-token dependencies otherwise lost under mean-field prediction.
- Engineered full-parameter post-training of a **32B MLLM** with Megatron-LM across data, tensor, pipeline, and sequence parallelism, with precision alignment. Evaluated mixtures of general-purpose and medical training data and varied positive-to-negative clinical case ratios.
- Built a multimodal agent with image-manipulation and DICOM tools, and trained its tool-use policy with reinforcement learning for medical image reasoning.
- Built a Chinese-language semantic reward with LLM-based NER and Qwen embeddings; added dynamic sigmoid calibration to prevent vanishing gradients and selected 16 high-signal trajectories from 128 rollouts for advantage estimation.

University of California, Santa Cruz, VLAA Lab

Mar. 2025 – Jul. 2025

Research Intern, Unified Multimodal Modeling

Advisor: Yuyin Zhou

- Co-designed **MeDiM**, a unified 7B discrete-diffusion MLLM that generates medical images, reports, or both with one shared backbone; adapted a pretrained MLLM and owned DeepSpeed training, improving downstream report generation by up to **31.6%**.
- Built an automated pipeline for seven visual conditions, including depth, segmentation, and edges; fused MM-DiT/U-Net features with CLIP semantics to strengthen MLLM spatial understanding.

Westlake University, AGI Lab

Oct. 2024 – Feb. 2025

Visiting Student, Controllable Generation

Advisor: Chi Zhang

- Proposed and led **Detail++**, a training-free framework that decomposes complex prompts into subject-specific sub-prompts and progressively injects attributes into their corresponding latent regions, reducing attribute leakage and style blending; improved attribute-binding performance by **15.9%** on SDXL, **11.1%** on SD3, and **16.3%** on Flux.
- Developed continual-learning personalization with concept-specific LoRAs and orthogonality regularization to reduce interference when composing independently learned concepts.